

house_prices

June 1, 2026

```
[85]: import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

```
[86]: df = pd.read_csv("house_prices.csv")
```

```
[87]: df.head(5)
```

```
[87]:
```

	Id	MSSubClass	MSZoning	LotFrontage	LotArea	Street	Alley	LotShape	\
0	1	60	RL	65.0	8450	Pave	NaN	Reg	
1	2	20	RL	80.0	9600	Pave	NaN	Reg	
2	3	60	RL	68.0	11250	Pave	NaN	IR1	
3	4	70	RL	60.0	9550	Pave	NaN	IR1	
4	5	60	RL	84.0	14260	Pave	NaN	IR1	

	LandContour	Utilities	...	PoolArea	PoolQC	Fence	MiscFeature	MiscVal	MoSold	\
0	Lvl	AllPub	...	0	NaN	NaN	NaN	0	2	
1	Lvl	AllPub	...	0	NaN	NaN	NaN	0	5	
2	Lvl	AllPub	...	0	NaN	NaN	NaN	0	9	
3	Lvl	AllPub	...	0	NaN	NaN	NaN	0	2	
4	Lvl	AllPub	...	0	NaN	NaN	NaN	0	12	

	YrSold	SaleType	SaleCondition	SalePrice
0	2008	WD	Normal	208500
1	2007	WD	Normal	181500
2	2008	WD	Normal	223500
3	2006	WD	Abnorml	140000
4	2008	WD	Normal	250000

[5 rows x 81 columns]

```
[88]: df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 1460 entries, 0 to 1459
Data columns (total 81 columns):
#   Column                Non-Null Count  Dtype
---  -
```

0	Id	1460	non-null	int64
1	MSSubClass	1460	non-null	int64
2	MSZoning	1460	non-null	str
3	LotFrontage	1201	non-null	float64
4	LotArea	1460	non-null	int64
5	Street	1460	non-null	str
6	Alley	91	non-null	str
7	LotShape	1460	non-null	str
8	LandContour	1460	non-null	str
9	Utilities	1460	non-null	str
10	LotConfig	1460	non-null	str
11	LandSlope	1460	non-null	str
12	Neighborhood	1460	non-null	str
13	Condition1	1460	non-null	str
14	Condition2	1460	non-null	str
15	BldgType	1460	non-null	str
16	HouseStyle	1460	non-null	str
17	OverallQual	1460	non-null	int64
18	OverallCond	1460	non-null	int64
19	YearBuilt	1460	non-null	int64
20	YearRemodAdd	1460	non-null	int64
21	RoofStyle	1460	non-null	str
22	RoofMatl	1460	non-null	str
23	Exterior1st	1460	non-null	str
24	Exterior2nd	1460	non-null	str
25	MasVnrType	588	non-null	str
26	MasVnrArea	1452	non-null	float64
27	ExterQual	1460	non-null	str
28	ExterCond	1460	non-null	str
29	Foundation	1460	non-null	str
30	BsmtQual	1423	non-null	str
31	BsmtCond	1423	non-null	str
32	BsmtExposure	1422	non-null	str
33	BsmtFinType1	1423	non-null	str
34	BsmtFinSF1	1460	non-null	int64
35	BsmtFinType2	1422	non-null	str
36	BsmtFinSF2	1460	non-null	int64
37	BsmtUnfSF	1460	non-null	int64
38	TotalBsmtSF	1460	non-null	int64
39	Heating	1460	non-null	str
40	HeatingQC	1460	non-null	str
41	CentralAir	1460	non-null	str
42	Electrical	1459	non-null	str
43	1stFlrSF	1460	non-null	int64
44	2ndFlrSF	1460	non-null	int64
45	LowQualFinSF	1460	non-null	int64
46	GrLivArea	1460	non-null	int64
47	BsmtFullBath	1460	non-null	int64

```

48 BsmthalfBath    1460 non-null    int64
49 FullBath        1460 non-null    int64
50 HalfBath        1460 non-null    int64
51 BedroomAbvGr   1460 non-null    int64
52 KitchenAbvGr   1460 non-null    int64
53 KitchenQual     1460 non-null    str
54 TotRmsAbvGrd   1460 non-null    int64
55 Functional      1460 non-null    str
56 Fireplaces      1460 non-null    int64
57 FireplaceQu     770 non-null     str
58 GarageType      1379 non-null    str
59 GarageYrBlt     1379 non-null    float64
60 GarageFinish    1379 non-null    str
61 GarageCars      1460 non-null    int64
62 GarageArea      1460 non-null    int64
63 GarageQual      1379 non-null    str
64 GarageCond      1379 non-null    str
65 PavedDrive      1460 non-null    str
66 WoodDeckSF      1460 non-null    int64
67 OpenPorchSF     1460 non-null    int64
68 EnclosedPorch   1460 non-null    int64
69 3SsnPorch       1460 non-null    int64
70 ScreenPorch     1460 non-null    int64
71 PoolArea        1460 non-null    int64
72 PoolQC          7 non-null       str
73 Fence           281 non-null     str
74 MiscFeature     54 non-null      str
75 MiscVal         1460 non-null    int64
76 MoSold          1460 non-null    int64
77 YrSold           1460 non-null    int64
78 SaleType        1460 non-null    str
79 SaleCondition   1460 non-null    str
80 SalePrice       1460 non-null    int64
dtypes: float64(3), int64(35), str(43)
memory usage: 924.0 KB

```

```
[89]: df.describe()
```

```

[89]:
count    1460.000000    1460.000000    1201.000000    1460.000000    1460.000000
mean       730.500000     56.897260     70.049958    10516.828082     6.099315
std       421.610009     42.300571     24.284752     9981.264932     1.382997
min         1.000000     20.000000     21.000000     1300.000000     1.000000
25%       365.750000     20.000000     59.000000     7553.500000     5.000000
50%       730.500000     50.000000     69.000000     9478.500000     6.000000
75%      1095.250000     70.000000     80.000000    11601.500000     7.000000
max      1460.000000    190.000000    313.000000    215245.000000    10.000000

```

	OverallCond	YearBuilt	YearRemodAdd	MasVnrArea	BsmtFinSF1	...	\
count	1460.000000	1460.000000	1460.000000	1452.000000	1460.000000	...	
mean	5.575342	1971.267808	1984.865753	103.685262	443.639726	...	
std	1.112799	30.202904	20.645407	181.066207	456.098091	...	
min	1.000000	1872.000000	1950.000000	0.000000	0.000000	...	
25%	5.000000	1954.000000	1967.000000	0.000000	0.000000	...	
50%	5.000000	1973.000000	1994.000000	0.000000	383.500000	...	
75%	6.000000	2000.000000	2004.000000	166.000000	712.250000	...	
max	9.000000	2010.000000	2010.000000	1600.000000	5644.000000	...	

	WoodDeckSF	OpenPorchSF	EnclosedPorch	3SsnPorch	ScreenPorch	...	\
count	1460.000000	1460.000000	1460.000000	1460.000000	1460.000000	...	
mean	94.244521	46.660274	21.954110	3.409589	15.060959	...	
std	125.338794	66.256028	61.119149	29.317331	55.757415	...	
min	0.000000	0.000000	0.000000	0.000000	0.000000	...	
25%	0.000000	0.000000	0.000000	0.000000	0.000000	...	
50%	0.000000	25.000000	0.000000	0.000000	0.000000	...	
75%	168.000000	68.000000	0.000000	0.000000	0.000000	...	
max	857.000000	547.000000	552.000000	508.000000	480.000000	...	

	PoolArea	MiscVal	MoSold	YrSold	SalePrice
count	1460.000000	1460.000000	1460.000000	1460.000000	1460.000000
mean	2.758904	43.489041	6.321918	2007.815753	180921.195890
std	40.177307	496.123024	2.703626	1.328095	79442.502883
min	0.000000	0.000000	1.000000	2006.000000	34900.000000
25%	0.000000	0.000000	5.000000	2007.000000	129975.000000
50%	0.000000	0.000000	6.000000	2008.000000	163000.000000
75%	0.000000	0.000000	8.000000	2009.000000	214000.000000
max	738.000000	15500.000000	12.000000	2010.000000	755000.000000

[8 rows x 38 columns]

```
[90]: df.isna().sum().sort_values(ascending=False)
```

```
[90]: PoolQC          1453
MiscFeature        1406
Alley              1369
Fence              1179
MasVnrType         872
...
MoSold             0
YrSold             0
SaleType           0
SaleCondition      0
SalePrice          0
Length: 81, dtype: int64
```

```
[91]: df = df.drop(["PoolQC", "MiscFeature", "Alley", "Fence", "MasVnrType",
↳ "FireplaceQu", "LotFrontage"], axis=1)
```

```
[101]: df = df.dropna(subset=["BsmtExposure", "GarageYrBlt", "MasVnrArea",
↳ "BsmtFinType2", "Electrical"])
```

```
[ ]: df["GarageYrBlt"] = pd.to_numeric(df["GarageYrBlt"], "coerce")
```

```
[ ]: df["GarageYrBlt"] = df["GarageYrBlt"].fillna(df["GarageYrBlt"].median())
```

```
[102]: df.isna().sum().sort_values(ascending=False)
```

```
[102]: Id                0
MSSubClass             0
MSZoning               0
LotArea               0
Street                0
..
MoSold                0
YrSold                0
SaleType              0
SaleCondition         0
SalePrice             0
Length: 74, dtype: int64
```

```
[103]: df_clean = df.copy
```

```
[105]: df.shape
```

```
[105]: (1338, 74)
```

```
[104]: df.info()
```

```
<class 'pandas.DataFrame'>
Index: 1338 entries, 0 to 1459
Data columns (total 74 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Id                    1338 non-null  int64
1   MSSubClass            1338 non-null  int64
2   MSZoning              1338 non-null  str
3   LotArea               1338 non-null  int64
4   Street                1338 non-null  str
5   LotShape              1338 non-null  str
6   LandContour           1338 non-null  str
7   Utilities             1338 non-null  str
8   LotConfig             1338 non-null  str
9   LandSlope             1338 non-null  str
```

10	Neighborhood	1338	non-null	str
11	Condition1	1338	non-null	str
12	Condition2	1338	non-null	str
13	BldgType	1338	non-null	str
14	HouseStyle	1338	non-null	str
15	OverallQual	1338	non-null	int64
16	OverallCond	1338	non-null	int64
17	YearBuilt	1338	non-null	int64
18	YearRemodAdd	1338	non-null	int64
19	RoofStyle	1338	non-null	str
20	RoofMatl	1338	non-null	str
21	Exterior1st	1338	non-null	str
22	Exterior2nd	1338	non-null	str
23	MasVnrArea	1338	non-null	float64
24	ExterQual	1338	non-null	str
25	ExterCond	1338	non-null	str
26	Foundation	1338	non-null	str
27	BsmtQual	1338	non-null	str
28	BsmtCond	1338	non-null	str
29	BsmtExposure	1338	non-null	str
30	BsmtFinType1	1338	non-null	str
31	BsmtFinSF1	1338	non-null	int64
32	BsmtFinType2	1338	non-null	str
33	BsmtFinSF2	1338	non-null	int64
34	BsmtUnfSF	1338	non-null	int64
35	TotalBsmtSF	1338	non-null	int64
36	Heating	1338	non-null	str
37	HeatingQC	1338	non-null	str
38	CentralAir	1338	non-null	str
39	Electrical	1338	non-null	str
40	1stFlrSF	1338	non-null	int64
41	2ndFlrSF	1338	non-null	int64
42	LowQualFinSF	1338	non-null	int64
43	GrLivArea	1338	non-null	int64
44	BsmtFullBath	1338	non-null	int64
45	BsmtHalfBath	1338	non-null	int64
46	FullBath	1338	non-null	int64
47	HalfBath	1338	non-null	int64
48	BedroomAbvGr	1338	non-null	int64
49	KitchenAbvGr	1338	non-null	int64
50	KitchenQual	1338	non-null	str
51	TotRmsAbvGrd	1338	non-null	int64
52	Functional	1338	non-null	str
53	Fireplaces	1338	non-null	int64
54	GarageType	1338	non-null	str
55	GarageYrBlt	1338	non-null	float64
56	GarageFinish	1338	non-null	str
57	GarageCars	1338	non-null	int64

```

58  GarageArea      1338 non-null  int64
59  GarageQual      1338 non-null  str
60  GarageCond      1338 non-null  str
61  PavedDrive      1338 non-null  str
62  WoodDeckSF      1338 non-null  int64
63  OpenPorchSF     1338 non-null  int64
64  EnclosedPorch   1338 non-null  int64
65  3SsnPorch       1338 non-null  int64
66  ScreenPorch     1338 non-null  int64
67  PoolArea        1338 non-null  int64
68  MiscVal         1338 non-null  int64
69  MoSold          1338 non-null  int64
70  YrSold          1338 non-null  int64
71  SaleType        1338 non-null  str
72  SaleCondition    1338 non-null  str
73  SalePrice       1338 non-null  int64

```

dtypes: float64(2), int64(35), str(37)

memory usage: 784.0 KB

```
[106]: df.describe()
```

```

[106]:
count      Id      MSSubClass      LotArea  OverallQual  OverallCond  \
count  1338.000000  1338.000000   1338.000000  1338.000000  1338.000000
mean    731.228700   56.136024  10706.294469    6.219731    5.596413
std     421.779691   41.252576  10336.621126    1.324472    1.078124
min       1.000000   20.000000   1300.000000    2.000000    2.000000
25%     366.250000   20.000000   7744.000000    5.000000    5.000000
50%     730.500000   50.000000   9600.000000    6.000000    5.000000
75%    1098.750000   70.000000  11760.750000    7.000000    6.000000
max    1460.000000  190.000000  215245.000000   10.000000    9.000000

      YearBuilt  YearRemodAdd  MasVnrArea  BsmtFinSF1  BsmtFinSF2  ...  \
count  1338.000000  1338.000000  1338.000000  1338.000000  1338.000000  ...
mean   1973.029148  1985.668909   110.360239   464.234679    49.218236  ...
std     29.563540   20.296463   185.604816   458.792420   166.196584  ...
min   1880.000000  1950.000000    0.000000    0.000000    0.000000  ...
25%   1956.000000  1968.000000    0.000000    0.000000    0.000000  ...
50%   1976.000000  1994.500000    0.000000   413.000000    0.000000  ...
75%   2001.000000  2004.000000   174.000000   733.000000    0.000000  ...
max   2010.000000  2010.000000  1600.000000  5644.000000  1474.000000  ...

      WoodDeckSF  OpenPorchSF  EnclosedPorch  3SsnPorch  ScreenPorch  \
count  1338.000000  1338.000000   1338.000000  1338.000000   1338.000000
mean     99.384903   47.782511    21.263827    3.585949   16.43423
std    127.537065   65.362562    60.843964   30.224622   58.05159
min      0.000000    0.000000    0.000000    0.000000    0.000000
25%      0.000000    0.000000    0.000000    0.000000    0.000000

```

50%	6.000000	28.000000	0.000000	0.000000	0.000000
75%	174.500000	70.000000	0.000000	0.000000	0.000000
max	857.000000	547.000000	552.000000	508.000000	480.000000

	PoolArea	MiscVal	MoSold	YrSold	SalePrice
count	1338.000000	1338.000000	1338.000000	1338.000000	1338.000000
mean	3.010463	42.932735	6.331839	2007.805680	186761.782511
std	41.961337	508.056255	2.699437	1.330691	78913.847668
min	0.000000	0.000000	1.000000	2006.000000	35311.000000
25%	0.000000	0.000000	5.000000	2007.000000	135000.000000
50%	0.000000	0.000000	6.000000	2008.000000	168500.000000
75%	0.000000	0.000000	8.000000	2009.000000	220000.000000
max	738.000000	15500.000000	12.000000	2010.000000	755000.000000

[8 rows x 37 columns]

```
[107]: df.isna().sum()
```

```
[107]: Id                0
       MSSubClass        0
       MSZoning          0
       LotArea           0
       Street            0
       ..
       MoSold            0
       YrSold            0
       SaleType          0
       SaleCondition      0
       SalePrice          0
       Length: 74, dtype: int64
```

```
[110]: df.duplicated().sum()
```

```
[110]: np.int64(0)
```